

Where does bat speed come from, and how much of it becomes exit velocity?

Predicting bat speed for hitters a model has never seen, and checking the lab's bat-ball physics against 2025 MLB data.

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Code: github.com/Andresperez397/obp-swing-speed

DATA

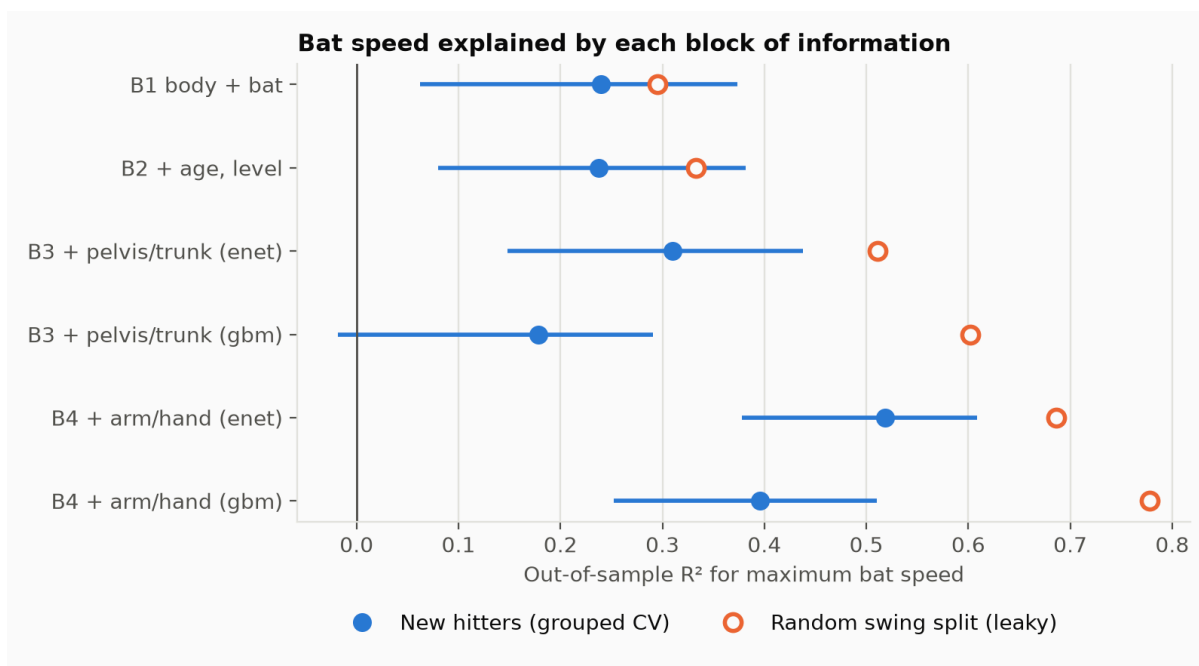
677 swings from 98 hitters in the OpenBiomechanics Project (Driveline Baseball): marker-based motion capture, about 120 swing metrics, exit velocity and HitTrax launch data, hitting off a machine at about 59 mph. The MLB comparison uses 118,793 balls in play with 2025 Statcast bat tracking.

APPROACH

- **Audit, then a frozen plan:** 22 redundant duplicate variables and 51 invalid pitch speeds found and handled before modeling; plan committed to git first.
- **Unseen hitters:** grouped CV by hitter, 10 folds × 20 repeats; leakage-tested.
- **Fair tuning:** elastic net and boosted trees both tuned by grouped inner CV.
- **Physics:** collision efficiency q from $EV = q \cdot v_{\text{pitch}} + (1+q) \cdot v_{\text{bat}}$ (Nathan 2003).

FINDINGS

- **Bat speed is a hitter trait:** 86% of its variance is between hitters (ICC 0.86).
- **Size gets a quarter of the way:** body and bat size explain R^2 0.24 for new hitters. Pelvis and trunk metrics add +0.06, not separable from zero; arm and hand metrics add +0.20.
- **Simple beat flexible:** tuned boosted trees trailed the elastic net. A leaky random split made them look best (0.78 vs 0.40).
- **Physics checks out:** between hitters, +1 mph bat speed = +1.22 mph exit velocity, matching $1 + q$. Within a hitter it's only +0.48, which may be contact quality or measurement noise.
- **Lab vs MLB:** $q = 0.236$ in the lab and 0.209 in MLB near squared-up contact. Both are near the textbook value of about 0.2.

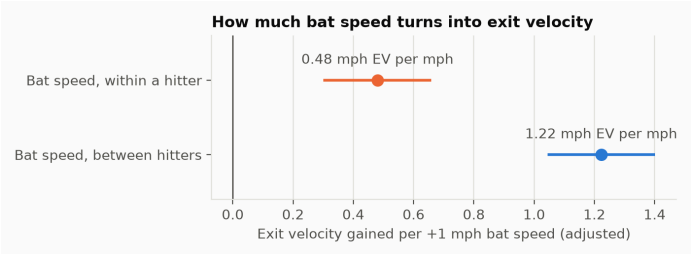


Out-of-sample R^2 for maximum bat speed by information block. Filled: validated on held-out hitters, with 95% CI. Hollow: the same model under a random swing split.

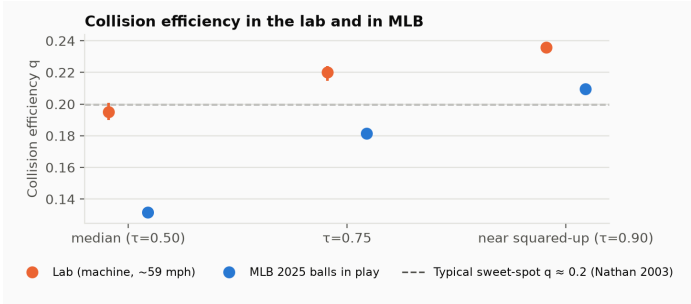
MODEL RESULTS (HELD-OUT HITTERS)

Model	R ²	95% CI	RMSE (mph)	R ² , leaky split
Body mass, height, bat size, handedness	0.24	0.06–0.37	4.28	0.30
+ age and playing level	0.24	0.08–0.38	4.28	0.33
+ 65 pelvis and trunk metrics (elastic net)	0.31	0.15–0.44	4.08	0.51
+ 27 arm and hand metrics (elastic net)	0.52	0.38–0.61	3.40	0.69
Same, tuned boosted trees	0.40	0.25–0.51	3.81	0.78

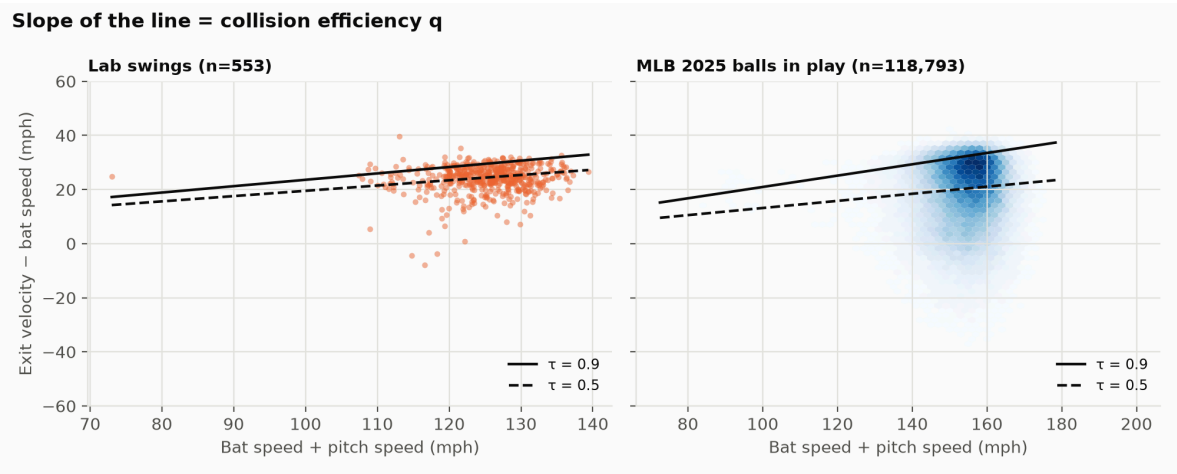
Mean-model RMSE 4.91 mph. R² and RMSE: averages over 20 repeats of 10-fold CV grouped by hitter. CIs: hitter bootstrap (2,000 resamples) of the first repeat.



Exit velocity per +1 mph of bat speed, adjusted for attack angle, body mass and handedness.



Collision efficiency q at three quantiles. MLB intervals are too narrow to see.



LIMITATIONS

The sample is 98 mostly college hitters at one facility, hitting a machine at about 59 mph, so results may not transfer directly to game swings. The lab bat's material isn't documented, and it affects collision efficiency. The MLB comparison is exploratory: different populations, pitch speeds and measurement systems. With 98 hitters, R² gains under about 0.1 aren't separable from zero.

Data: The OpenBiomechanics Project, Driveline Baseball (CC BY-NC-SA 4.0); Baseball Savant / Statcast (MLB Advanced Media). A derived work under the same license, not affiliated with or endorsed by Driveline or MLB.